RoadGP: Automating the Diagnosis and Treatment of Road Defects

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Introduction

We present **RoadGP**, a new web-based component-level highway diagnostics tool developed in consultation with National Highways (NH) pavement experts that integrates pavement defect, repair, and causation information from previous decision support systems (DSSs) [1,2], as well as new concrete roads-specific knowledge extracted from NH manuals [3].

This tool is inspired by the WebMD symptom checker, where patients provide information about their symptoms, and the tool produces possible diagnoses and appropriate treatment plans. Similarly, RoadGP takes information on a given road section, along with any defects present as an input, and outputs:

1. Probable causes for the defects, and their associated preventative treatments.
2. Recommended repair strategies.

While stand-alone, RoadGP can be integrated into a semi-autonomous highway maintenance pipeline to improve efficiency.

Overall Pipeline

RoadGP fits into a larger microservice-based architecture made to support semi-autonomous road maintenance planning, decision-making, and implementation. This architecture includes a modular framework (see Figure 1) that ensures RoadGP is both autonomous and collaborative. This also allows for ongoing extension, where future services can be introduced without restructuring the core system.

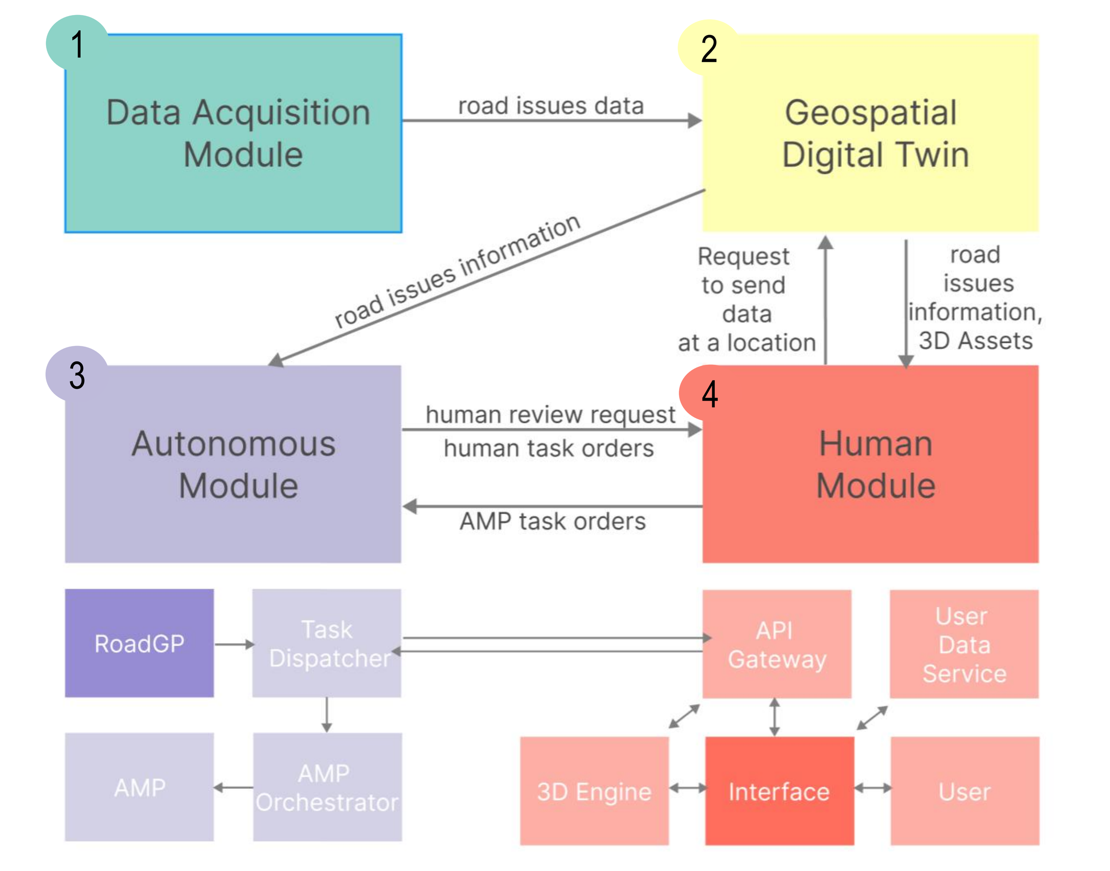

Figure 1: Complete semi-autonomous highway maintenance architecture. RoadGP sits inside the Autonomous Module, communicating with the other components to take defect data as an input and output task orders to an Autonomous Maintenance Plant (AMP) or for human review.

The **Data Acquisition** (DA) module collects road condition data originating from multiple sources, including processed data from connected vehicles, historical datasets, traffic patterns, and accident frequency.

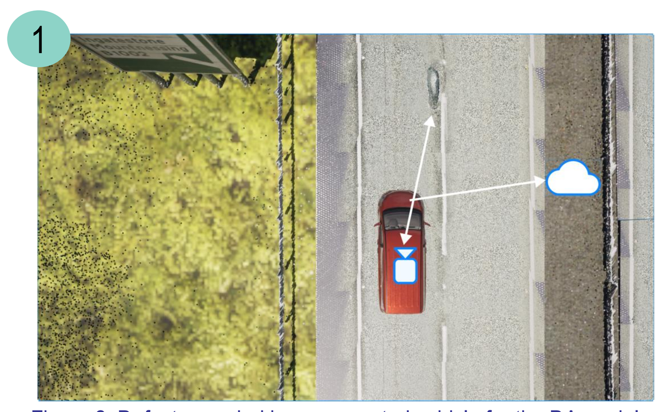

Figure 2: Defect recorded by a connected vehicle for the DA module

Figure 3: Map in the GDT displaying the newly recorded defect.

The **Geospatial Digital Twin** (GDT) module organises data received from the DA module within an existing Digital Twin (DT) system: A 2D map-based DT interface, a geospatial PostgreSQL database, and a 3D geometry repository that stores tiled mesh data.

RoadGP is deployed in the **Autonomous Module** (AM), receiving pre-processed defect data (using AI-powered computer vision, or manual human review) from the GDT module. It acts as a decision support service, automatically triggering repair tasks with recommended treatment and repair strategies.

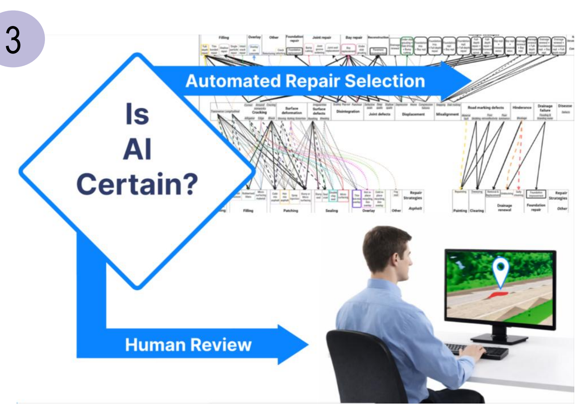

Figure 4: Processing of new defect data, either by RoadGP in the AM, or reviewed by a human in the HM.

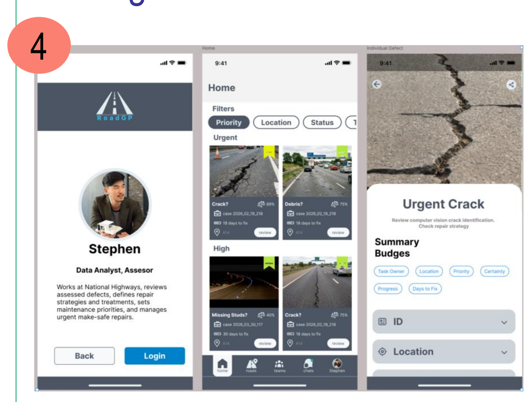

Figure 5: HM interface for viewing RoadGP outputs and managing various task orders.

The **Human Module** (HM) is the primary interface layer for users. It includes a web dashboard, a 3D virtual interface, and a notification system. Human users can view recommendations from RoadGP, approve or modify tasks, and provide feedback that informs the autonomous decision cycle.

Methodology: Knowledge Gathering

The core of developing RoadGP is understanding how to address and prevent highway defects. To organise the DSS, highway **defects** are first grouped by disease, and classified by **severity**, then all relevant **repair** strategies and defect **causes**, along with preventative **treatment** plans, are found.

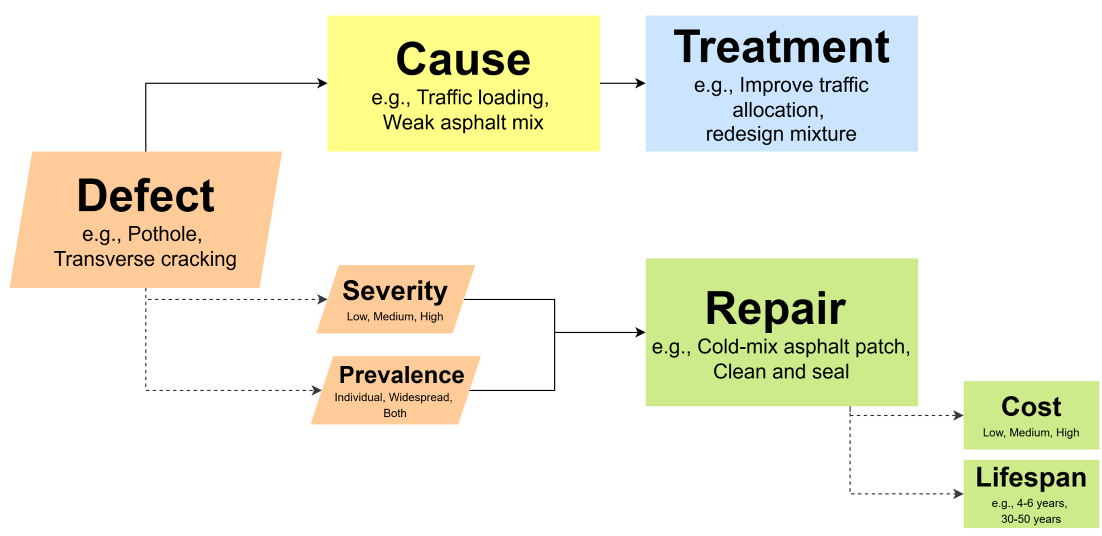

Figure 6: Links between inputs (defects, their severity and prevalence), and outputs (cause and treatment, and repair) in the DSS. Each repair strategy also has an associated cost and lifespan.

To improve decision-making, additional criteria are considered:

- **Lifespan**: the expected period of time that a given repair will last for before failing.
- **Cost**: the financial cost of carrying out the repair. Since the exact cost information for repairs is generally unavailable, rough estimates were inferred from existing literature.
- **Prevalence**: some repairs are better suited for situations where there are widespread instances of defects across a road section over single individual defect instances.

Most of this knowledge, summarised in Figure 6, was obtained by investigating maintenance manuals, papers, and online guides [3-6] for every repair strategy, and consulting industry specialists.

Methodology: Network Design

Causes and Treatment Plans

To determine what is likely to cause defects on a given road section, the DSS network is designed such that, when multiple defects are entered, all connections (see Figure 7) are assessed, and the most frequent cause is displayed as the most likely, along with any other relevant alternatives. A second, direct connection layer is also formed mapping causes to their treatment plans.

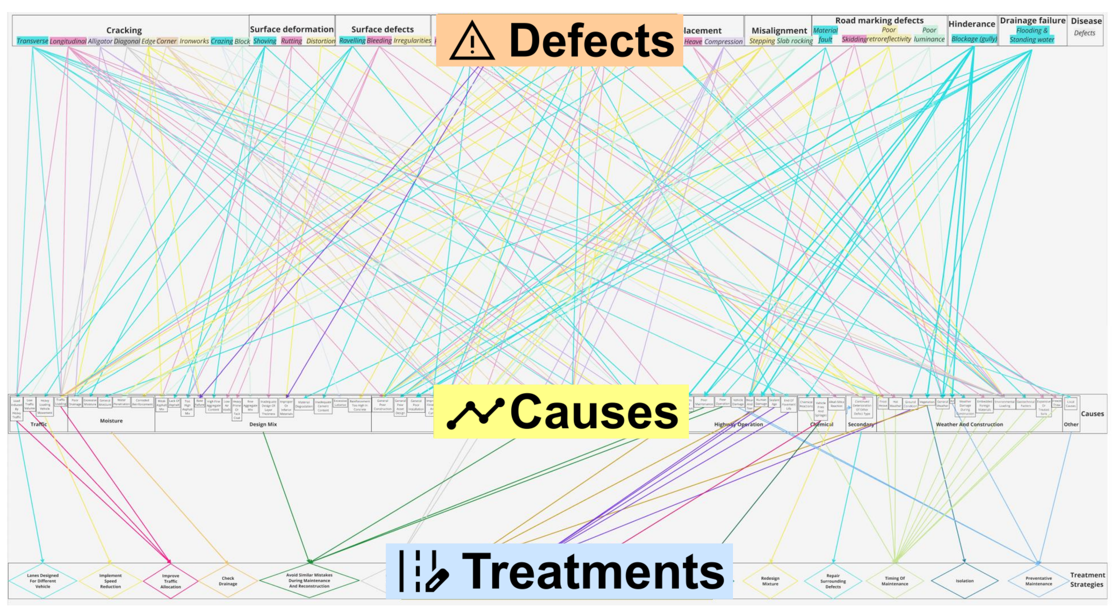

Figure 7: Complex relationships mapped between defects and causes, and between causes and their corresponding treatments, as employed by RoadGP to suggest probable causes and recommend treatment plans.

Repair Strategies

To decide what repair is most suitable; their prevalence, severity, and the surface material type affect what strategies are available. Once all the connections have been established (see Figure 8), the repair strategies with the most connections to the selected set of defects are considered for further implementation.

Prioritisation and tie-breakers then occur in the following order before recommending the most relevant repair:

1. **Cost**: 'Low', then 'Medium', then 'High' cost repairs.
2. **Average lifespan**: longer-lasting repairs are privileged.
3. **Lifespan range**: shorter ranges within treatment plans mean more consistently lasting repairs.

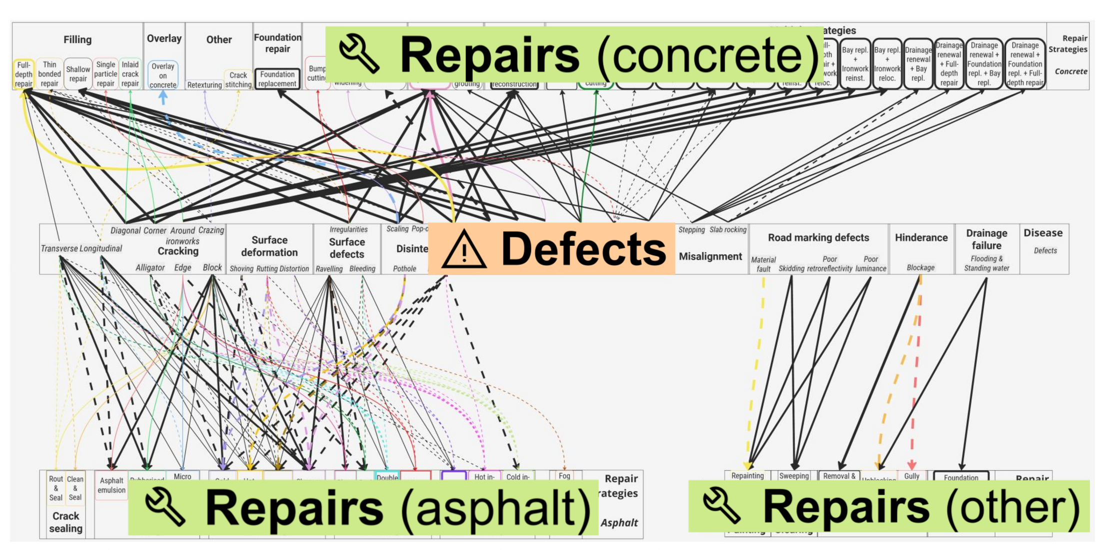

Figure 8: Complex relationships mapped between defects and repairs (separated between asphalt roads, concrete roads, and other assets such as drainage and road markings), as employed by RoadGP to recommend repair strategies.

Network Implementation

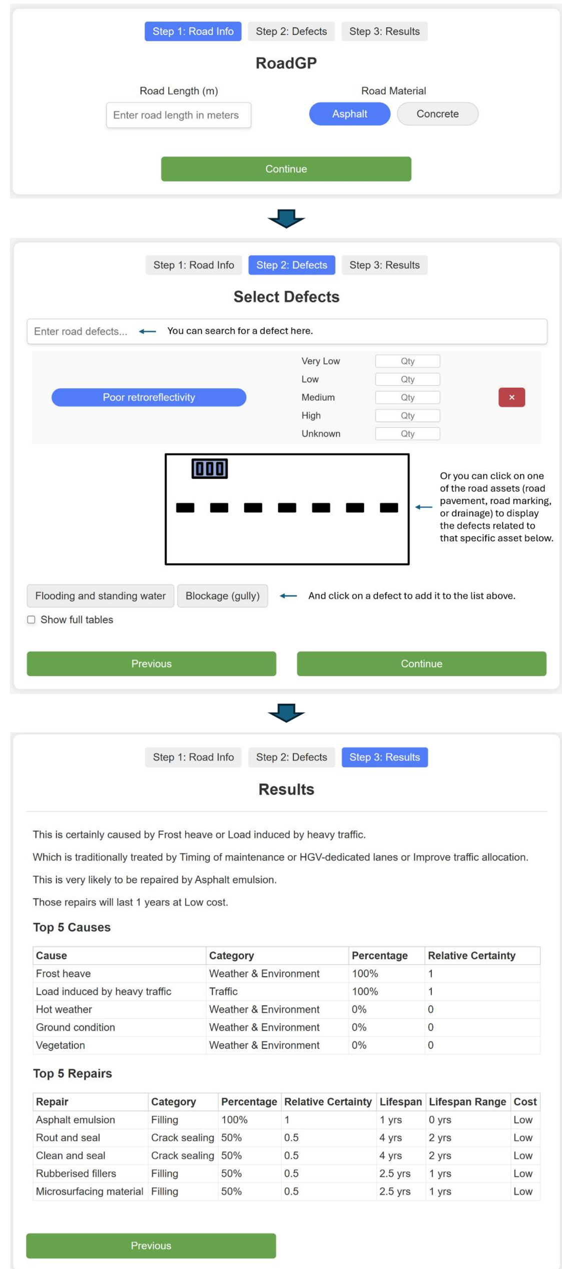

Figure 9: RoadGP web interface: road information input (top), defect selection and severity entry (middle), and diagnostic results with cause and repair strategies (bottom).

RoadGP is implemented as a client-side web application that can be used directly from a wide range of commonly available devices. This implementation integrates four pre-processed data tables:

- Defect–cause associations;
- Defect–repair associations;
- Repair strategies with cost and lifespan attributes;
- Cause-treatment associations.

The interface follows a three-stage interaction model illustrated in Figure 9. Each selected defect–severity–prevalence input is mapped to candidate causes and repair vectors. The system accumulates frequencies across all inputs, before performing prioritisation. Results are presented both in natural language and tabular form.

The RoadGP web interface can be tested by scanning the following QR code. Further project details are also available.

Conclusion and Future Work

We propose a novel tool for automating highway defect diagnoses at the component level by implementing a comprehensive DSS that recommends repair strategies, possible causes, and their treatments. RoadGP can be embedded within an accessible and interpretable web interface, and can work independently, or within a larger architecture that streamlines semi-autonomous decision making and maintenance planning.

It is important to note that RoadGP remains simple and deterministic, and future versions will account for more intricacies within the maintenance process. Data-driven methods are being developed, that reflect which strategies have historically been favoured, using textual repair data from NH, processed via LLM. Data-driven weighting of defect causes will also be explored via statistical spatial modelling. The updated tool will also undergo quantitative testing in consultation with pavement experts and contractors.

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